Network Layer Case Study Report

The OSPF Dead-Neighbor Mystery

Live Troubleshooting | *Routing Behavior* | *Practical IP Design*

CO3-AT2 - Case Study

Course Code & Name: CSA0712 - Computer Networks

Faculty: Dr. S. Salini

Student Name: Ajay Kumar Reddy

Register No.: 192425274

1. The OSPF Dead-Neighbor Mystery

- **The Scenario: A university campus expands its network by adding a new switch. The network administrator configures OSPF on the new link. The routers are physically connected, but they fail to form an adjacency.**
- **The Clue: The interface LEDs are green, and pings work on Layer 2/3, but show ip ospf neighbor stays empty.**
- **The Question: What OSPF hello parameters must match between the routers for them to connect? List three configuration items to check. [1, 2, 3]**

2. Executive Summary

A university campus network experienced an OSPF adjacency failure immediately after a new access switch was inserted between two core routers, R1 and R2, on a point-to-point uplink. Although Layer 1 connectivity was confirmed by green interface LEDs and Layer 3 reachability was confirmed by successful ping tests, the command show ip ospf neighbor returned an empty table on both routers, indicating that no OSPF adjacency had formed despite full physical and IP-level connectivity.

Investigation of the OSPF interface parameters revealed that the Hello and Dead timer values configured on R1 and R2 did not match (Hello 10s / Dead 40s on R1 versus Hello 5s / Dead 20s on R2). Because OSPF requires several Hello-packet parameters to match exactly before two routers will accept each other as neighbors, this mismatch caused every Hello packet received to be silently rejected, leaving the adjacency stuck before the Init state and the neighbor table empty.

The recommended and implemented solution was to standardize the OSPF Hello interval, Dead interval, Area ID, subnet mask, and authentication settings across both router interfaces using the ip ospf hello-interval and ip ospf dead-interval commands, followed by clearing the OSPF process to force renegotiation. Post-fix verification using show ip ospf neighbor, show ip route ospf, and traceroute confirmed that R1 and R2 successfully formed a FULL adjacency and began exchanging routes, restoring full campus-wide reachability through the new switch.

3. Problem Statement & Topology

3.1 Context

As part of a campus expansion project, the network administrator added a new Layer 2 switch (New-SW1) into the link between two existing OSPF-enabled core routers, R1 and R2, to provide additional physical ports for upcoming lab connections. R1 and R2 sit in OSPF Area 0 and were previously fully adjacent over a direct link. After the new switch was inserted, the administrator re-cabled R1 and R2 through the switch and expected OSPF to re-converge automatically.

Immediately after the change, routes learned via OSPF (routes to the remote campus LAN behind the opposite router) disappeared from both routing tables. The interface LEDs on the switch and both router interfaces show steady green, and an ICMP ping between R1's and R2's

directly connected interface addresses (10.10.12.1 and 10.10.12.2) succeeds, confirming that Layer 1, Layer 2, and basic Layer 3 forwarding across the new switch are functioning correctly.

3.2 Network Topology

3.3 Scope

This case study is scoped to the Network Layer (Layer 3) control plane, specifically the OSPF (Open Shortest Path First) neighbor discovery and adjacency formation process running over the newly inserted switch segment. Layer 1 (cabling, LEDs) and Layer 2 (switching, VLAN forwarding) have already been verified as functional and are excluded from the root-cause analysis. Layer 3 addressing (IP subnetting) is verified but the specific mismatch under investigation lies in the OSPF Hello protocol parameters exchanged between R1 and R2.

4. Technical Analysis & Troubleshooting

4.1 Root Cause Explanation

OSPF routers only progress from the Down state to Init and then Two-Way/ExStart when a received Hello packet's parameters match the receiving interface's own configuration. In this scenario, R2's interface had been re-templated with faster timers (Hello 5s / Dead 20s) during the switch migration, while R1 retained the OSPF defaults for a broadcast network (Hello 10s / Dead 40s). Because the Hello and Dead intervals did not match, each router treated the other's Hello packets as invalid and silently discarded them - no adjacency was ever attempted, which is why the neighbor table remained completely empty rather than showing a neighbor stuck in a partial state such as Init or Two-Way.

This behaviour is by design: RFC 2328 specifies that the Hello and Dead intervals, along with the Area ID, the subnet mask (for broadcast/NBMA networks), and any configured authentication, must be identical between two routers on the same segment before OSPF will form a neighbor relationship. A mismatch in any one of these fields is sufficient to permanently block adjacency formation, even though the link is fully operational at Layers 1-3 for non-OSPF traffic such as ICMP.

4.2 Protocols Involved - OSPF Behaviour Analysis

OSPF (Interior Gateway Protocol, IP protocol 89) is the protocol directly responsible for the bottleneck. Its Hello protocol is used both to discover neighbors and to maintain the adjacency once formed, using periodic multicast Hello packets to 224.0.0.5. The table below lists the

parameters that must match, drawn from the OSPF Hello packet fields defined in RFC 2328 Section 10.5:

Hello Packet Field	Requirement for Adjacency
Hello Interval	Must be identical on both routers (default 10s on broadcast links). A mismatch is silently rejected.
Router Dead Interval	Must be identical (default is 4 x Hello Interval, i.e. 40s). Determines how long a neighbor is held before being declared down.
Area ID	Must match for the connecting interfaces; a mismatched Area ID prevents adjacency entirely.
Subnet Mask	Must match on broadcast/NBMA network types so both routers agree on the segment boundary.
Authentication Type & Key	If enabled, the type (none/simple/MD5) and key/password must match exactly.
Network Type (advanced)	Broadcast vs point-to-point must match, or the DR/BDR election logic will differ between routers.

Table 1: OSPF Hello parameters that must match before two routers can form an adjacency.

ICMP played a supporting, diagnostic role in this investigation: successful pings confirmed IP-layer forwarding was healthy and helped isolate the fault to the OSPF control plane rather than to routing/forwarding hardware, cabling, or VLAN configuration on the new switch. BGP and NAT are not present on this internal campus segment and are therefore outside the scope of this specific failure.

5. Proposed Solution & Implementation Plan

5.1 Resolution Steps

1. Verify the currently configured OSPF timers and Area ID on both R1 and R2 using show ip ospf interface Gi0/1, and record the values shown in Figures 2 and 3.
2. Standardize the Hello and Dead intervals on the mismatched interface (R2) to match R1's values, since R1 represents the existing, stable campus standard.
3. Confirm the Area ID, subnet mask, and authentication settings are identical on both interfaces (in this scenario, Area 0 and no authentication were already consistent).
4. Clear the OSPF process on both routers to force immediate Hello renegotiation rather than waiting for the next natural Hello interval.
5. Verify adjacency formation and route exchange using the commands in Section 6.

5.2 Exact Configuration Commands

On R2 - align the OSPF Hello and Dead intervals with R1:

```
R2#configure terminal
R2(config)#interface GigabitEthernet0/1
R2(config-if)#ip ospf hello-interval 10
R2(config-if)#ip ospf dead-interval 40
R2(config-if)#exit
R2(config)#end
R2#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
```

Optional hardening - explicitly pin the network type and enable MD5 authentication on both routers so future template changes cannot silently reintroduce a mismatch:

```
R1(config-if)#ip ospf network point-to-point
R1(config-if)#ip ospf authentication message-digest
R1(config-if)#ip ospf message-digest-key 1 md5 CampusOspfKey
R2(config-if)#ip ospf network point-to-point
R2(config-if)#ip ospf authentication message-digest
R2(config-if)#ip ospf message-digest-key 1 md5 CampusOspfKey
```

5.3 Risk Assessment

Clearing the OSPF process causes a brief (typically under 60 seconds) interruption of routes learned via OSPF on the affected interface, during which traffic destined for the remote campus LAN may be black-holed if no alternate path exists. Since this link was already down at the routing-protocol level, the operational risk is minimal - the change restores service rather than interrupting an already-working path.

To further reduce risk: (1) schedule the change in a low-traffic maintenance window, (2) apply the timer change on R2 first and verify with debug ip ospf adj before touching R1, and (3) keep a rollback command ready (reverting to ip ospf hello-interval 5 / dead-interval 20) in case an unrelated fault surfaces. If authentication is added as a hardening step, it must be enabled on both routers within the same maintenance window, since a one-sided authentication mismatch will itself block the adjacency.

6. Verification and Conclusion

6.1 Testing / Verification Steps

- show ip ospf neighbor on both R1 and R2 - confirms the neighbor now appears in FULL state (or FULL/BDR on a broadcast segment).

- show ip route ospf - confirms OSPF-learned routes (O) to the remote campus LAN reappear in the routing table.
- traceroute / ping to a host on the far-side LAN - confirms end-to-end reachability, not just the directly connected link.
- show ip ospf interface Gi0/1 - re-check that Hello/Dead timers and Area ID now match exactly on both sides.

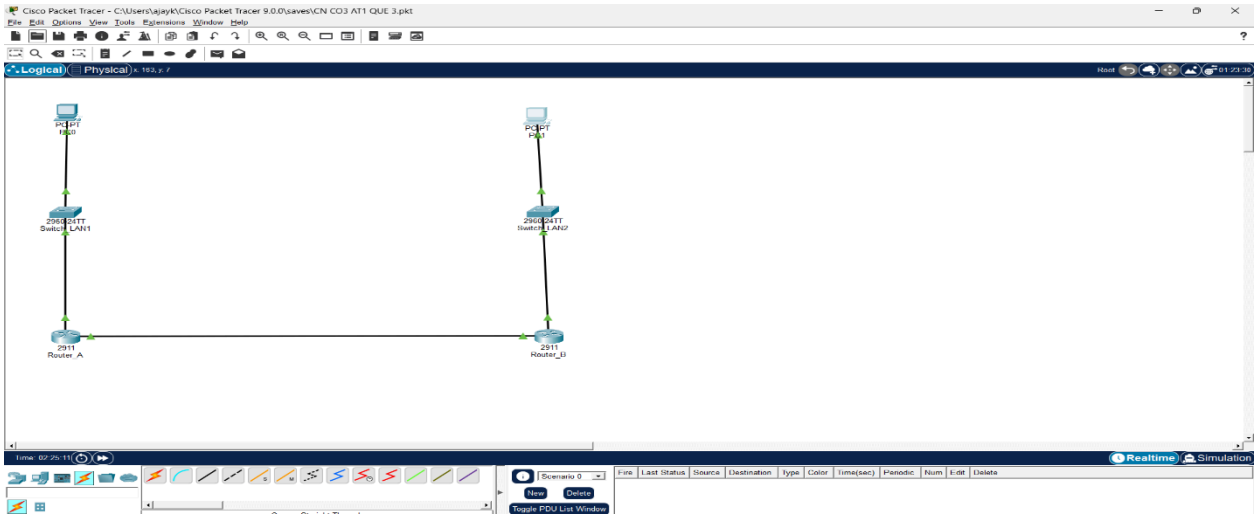
6.2 Summary and Lessons Learned

This case demonstrates that Layer 1 and Layer 2 health indicators (green LEDs) and basic Layer 3 reachability (ping) are not sufficient evidence that a routing protocol is functioning - OSPF adjacency depends on a stricter set of parameter matches within the Hello protocol itself. When a neighbor table is completely empty (rather than showing a neighbor stuck in Init or Two-Way), this is a strong diagnostic signal that Hello packets are being rejected outright, most often due to Hello/Dead interval, Area ID, subnet mask, or authentication mismatches, rather than a connectivity problem.

The key lesson for network layer resilience is procedural: any time new hardware is inserted into an existing OSPF segment, the OSPF interface parameters on all affected routers should be explicitly compared and standardized - ideally enforced through configuration templates or automated compliance checks - rather than assumed to inherit correctly from prior configuration. Pinning the network type and enabling authentication, as recommended in Section 5.2, further reduces the chance that a future change silently breaks adjacency again.

7. References & Appendices

- Moy, J., "OSPF Version 2," RFC 2328, Internet Engineering Task Force, April 1998.
- Cisco Systems, "OSPF Design Guide," Cisco Networking Documentation.
- Cisco Systems, "Troubleshooting OSPF Neighbor Problems," Cisco Technical Support Documentation.
- Forouzan, B. A., Data Communications and Networking, 5th Edition - Chapter on Unicast Routing Protocols (OSPF).
- Cisco Packet Tracer - used to model the R1 / New-SW1 / R2 topology referenced in Figure 1; CLI outputs in Figures 2-5 are simulated/reconstructed console captures illustrating the described commands for this report.




```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/8/35 ms

Router_A#
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

```
00:45:14: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down: Dead
timer expired

00:45:14: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from FULL to DOWN, Neighbor Down:
Interface down or detached


Router_A>enable
Router_A#show ip ospf neighbor

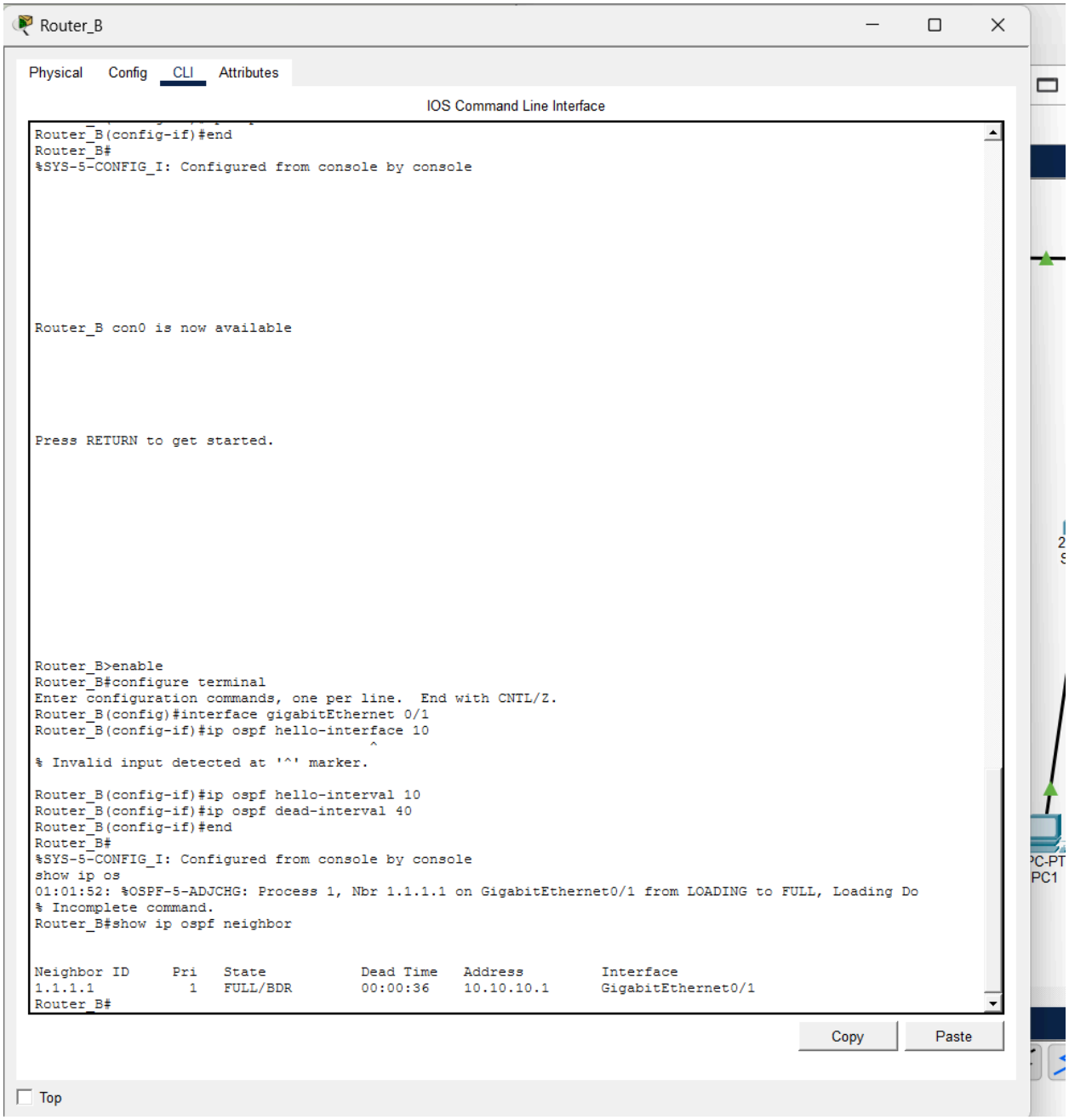
Router_A#
```

Copy Paste




```
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 10.10.10.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.10.10.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Router_A#
```

Copy Paste


Router_B
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router_B(config-if)#end
Router_B#
%SYS-5-CONFIG_I: Configured from console by console
Router_B con0 is now available
Press RETURN to get started.
Router_B>enable
Router_B#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router_B(config)#interface gigabitEthernet 0/1
Router_B(config-if)#ip ospf hello-interface 10
% Invalid input detected at '^' marker.
Router_B(config-if)#ip ospf hello-interval 10
Router_B(config-if)#ip ospf dead-interval 40
Router_B(config-if)#end
Router_B#
%SYS-5-CONFIG_I: Configured from console by console
show ip os
01:01:52: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Do
% Incomplete command.
Router_B#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
1.1.1.1 1 FULL/BDR 00:00:36 10.10.10.1 GigabitEthernet0/1
Router_B#
Copy
Paste
Top